MANUSCRIPT

LPT: Orchestrating bounded agentic loops for automated protein binder design

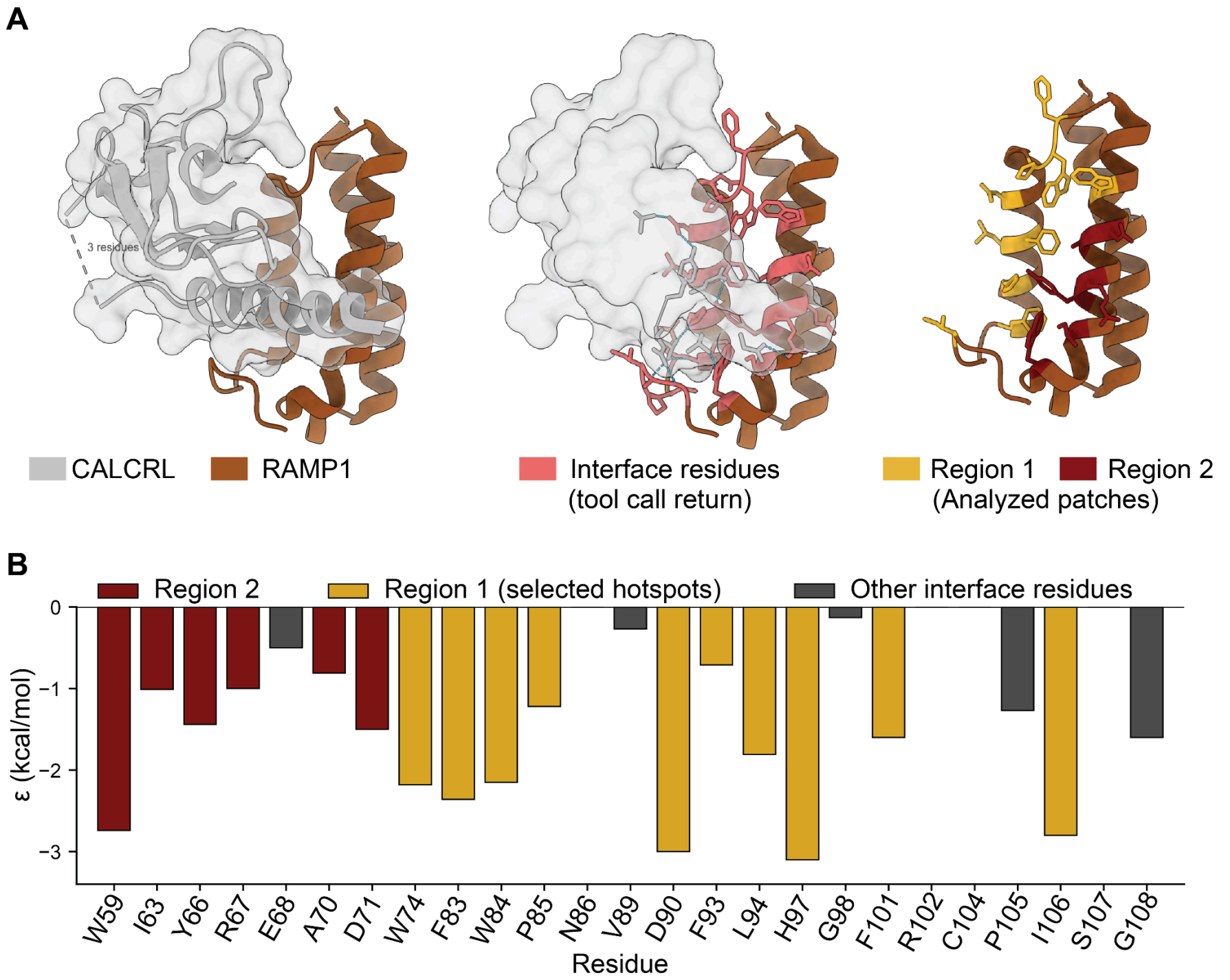
Fixed orchestration of LLM stages, each running a bounded agentic loop with its own system prompt and tool access, backed by a curated literature database. One natural-language prompt in; ranked, in silico validated binder designs out.

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Little Protein Tiger (LPT) · github.com/MichaUckelmann/little-protein-tiger

FIGURE 3 / STRUCTURE ANALYSIS

“Design de novo inhibitors for pain receptors that are promising drug targets.”

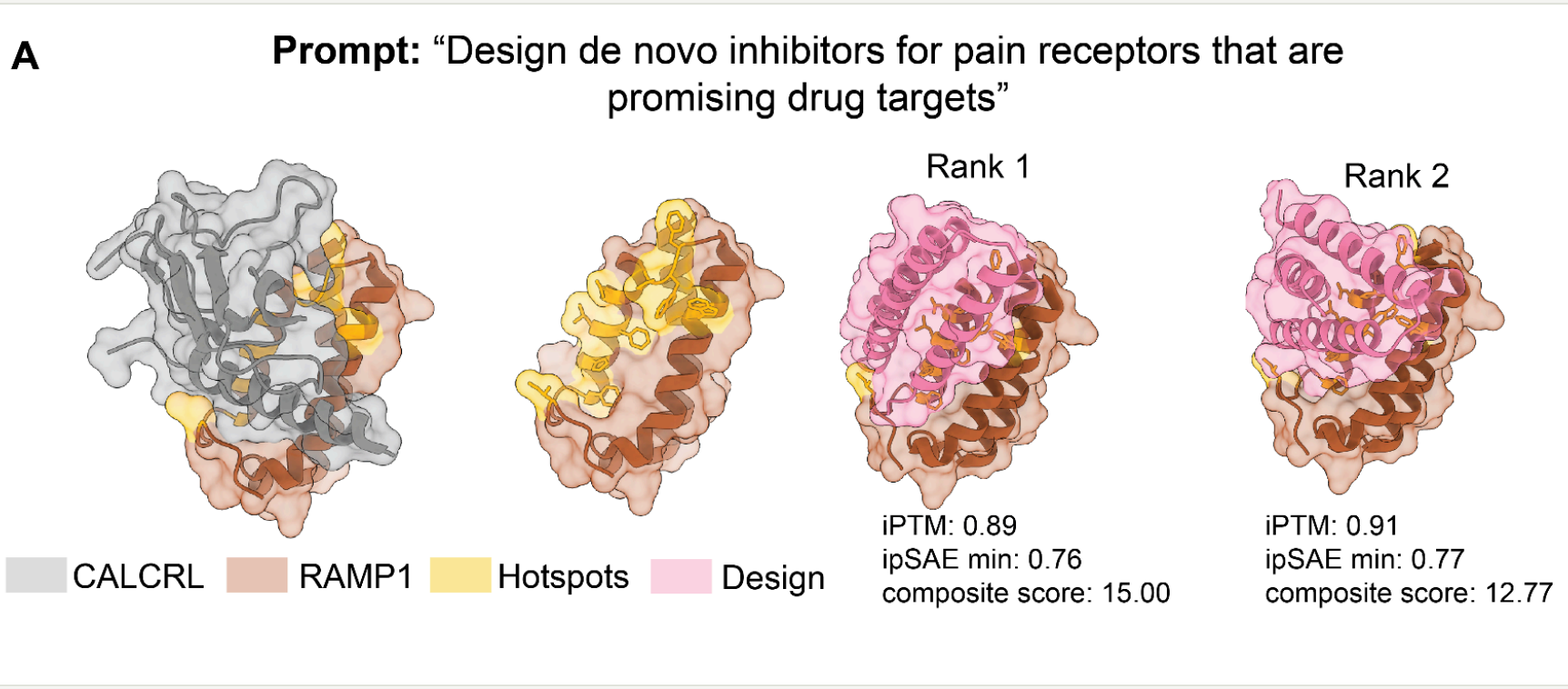


FROM PROMPT TO HOTSPOTS

Starting from the prompt alone, the pipeline selected CALCRL–RAMP1 (PDB 3N7S). The structure-analysis tool returns the interface analysis: interface residues, H-bonds and an estimated binding contribution ε per residue. The model then groups residues into clusters and chooses Region 1, a 10-residue patch spanning the core heterodimerisation surface of RAMP1, and forwards it to the design phase as hotspots.

FIGURE 4A / END-TO-END RUN 1

Campaign results

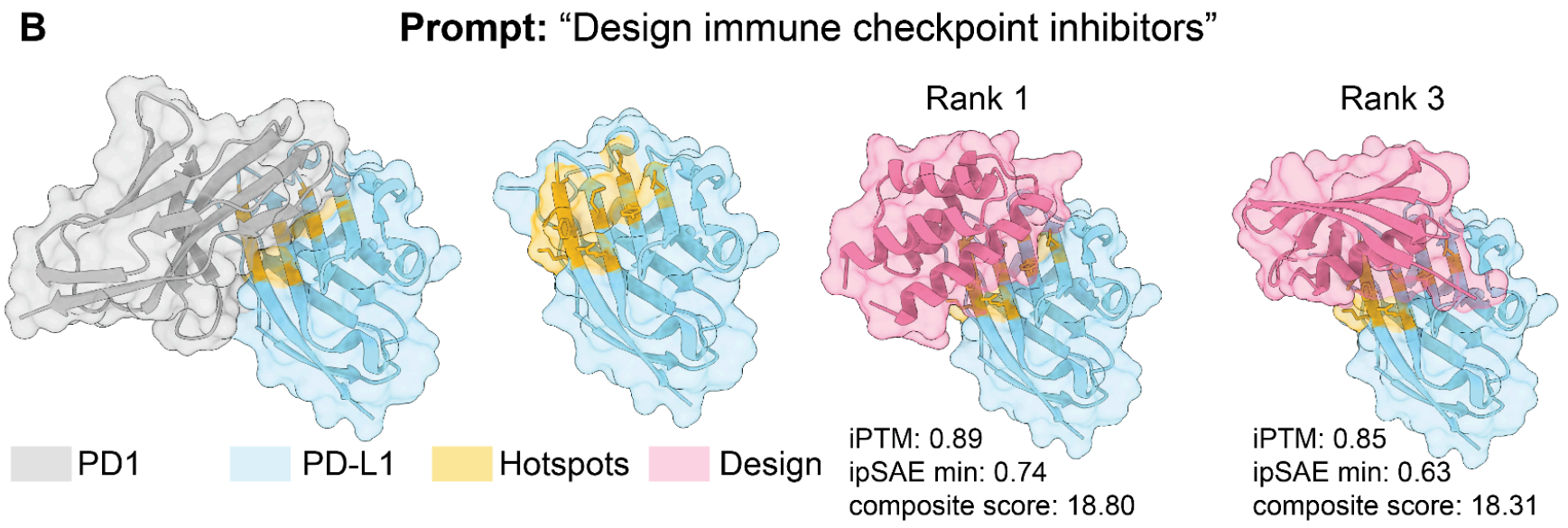


The pipeline chose to disrupt CALCRL–RAMP1 (PDB 3N7S). Hotspots in gold; two highly ranked designs in pink, with ipTM, minimum ipSAE and composite score.

17.9%	$0.33	~11 h
OF REFOLDS PASS ALL GATES, IPTM ≥ 0.7	LLM API COST	GPU TIME, LOCAL WORKSTATION

FIGURE 4B / END-TO-END RUN 2

“Design immune checkpoint inhibitors.”



The pipeline chose to disrupt PD-1/PD-L1 (PDB 4ZQK) and designed binders to PD-L1. Hotspots in gold; two highly ranked designs in pink, with ipTM, minimum ipSAE and composite score.

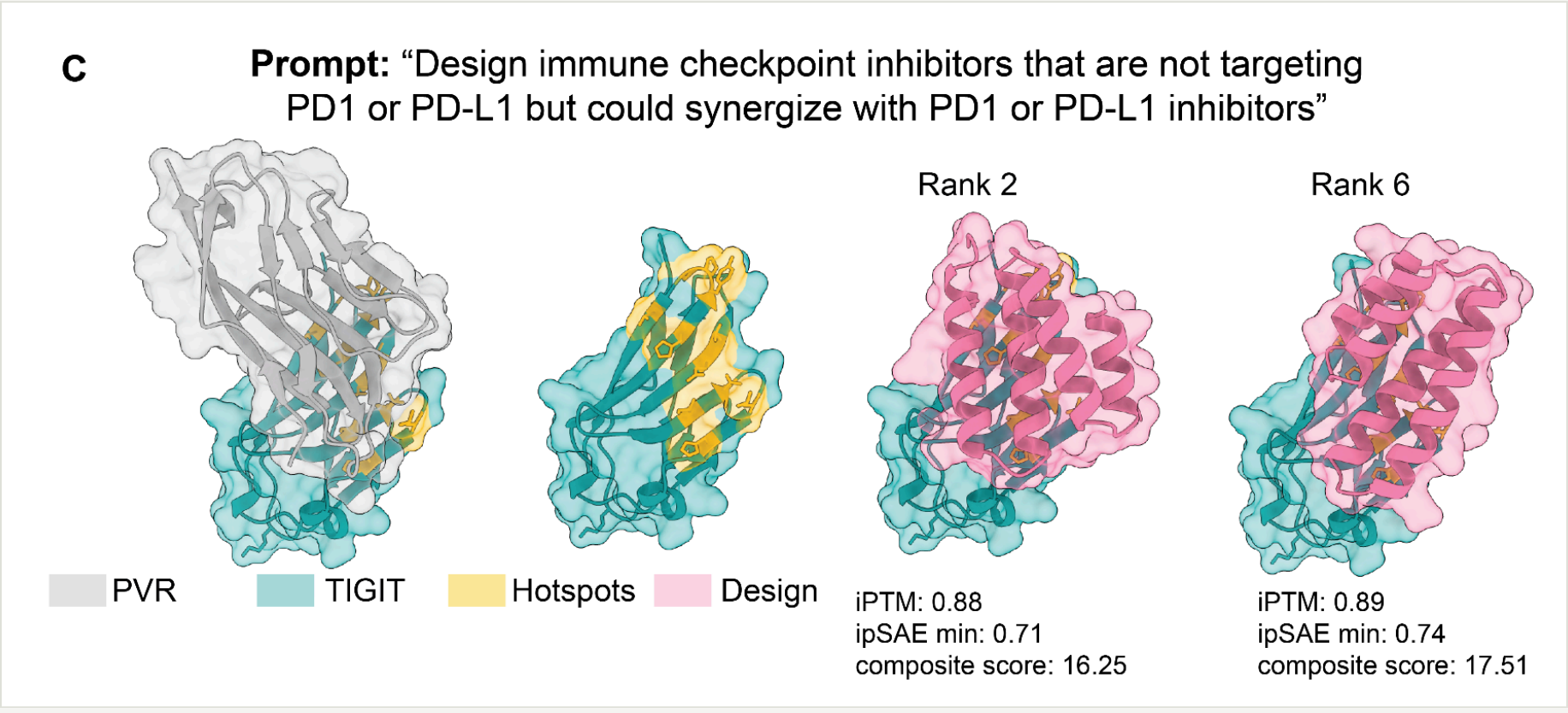
9.6%
OF REFOLDS PASS ALL GATES, IPTM ≥ 0.7

$0.42
LLM API COST

~34 h
GPU TIME, LOCAL WORKSTATION

FIGURE 4C / END-TO-END RUN 3

“...checkpoint inhibitors that are not targeting PD1 or PD-L1 but could synergize with them.”

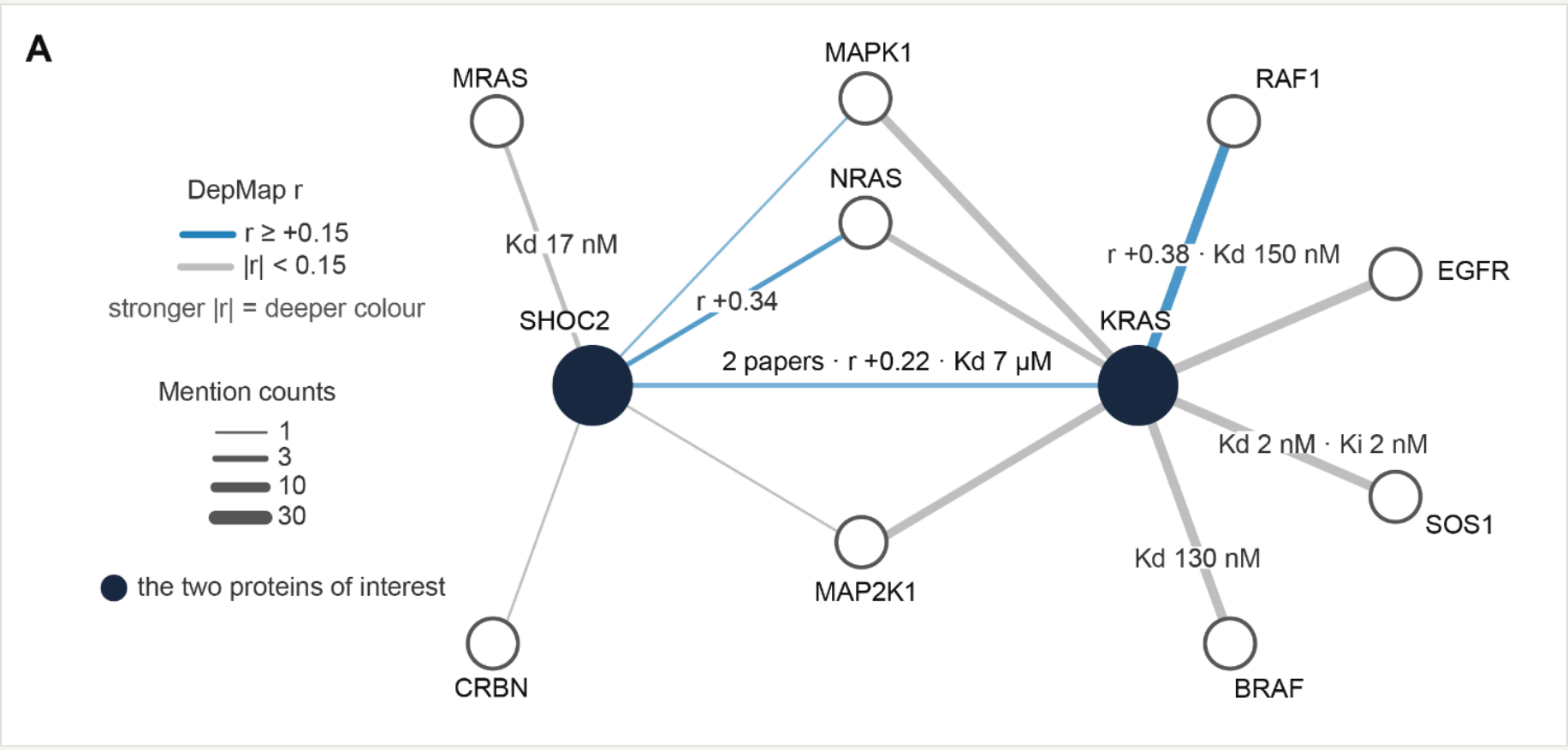


The pipeline chose to disrupt the TIGIT–PVR (CD155) interaction (PDB 3UDW). Hotspots in gold; two highly ranked designs in pink, with ipTM, minimum ipSAE and composite score.

10%	$0.39	~20 h
OF REFOLDS PASS ALL GATES, IPTM ≥ 0.7	LLM API COST	GPU TIME, LOCAL WORKSTATION

FIGURE 2A / LITERATURE DATABASE

~14,000 papers, digested and linked into an interaction graph.



Nodes are proteins; edges carry affinity information and the number of papers mentioning the interaction (edge width). Edge colour marks the DepMap CRISPR gene-effect correlation of the pair. The graph shown is a small subgraph.

CLOSED-BOOK AGAINST OPEN-BOOK EXAM

48 questions answerable from the database. With database access: 41 correct, 7 “unknown”. Without: 0 correct, 47 “unknown”, 1 wrong. The exam was biased towards database content and some questions were leading.

LITERATURE LAYER

Each paper is digested into a structured fingerprint.

Key findings with evidence, source span and confidence; proteins and structures; pathway context. Example (abridged):

USP26 functions as a negative regulator of cellular reprogramming by stabilising PRC1 complex components

DOI 10.1038/s41467-017-00301-4 · pathway_biology · experimental_in_vitro · schema 2.0

CONTEXT HOOK

In mouse embryonic fibroblasts undergoing OSKM-mediated reprogramming, the deubiquitinase USP26 physically interacts with CBX4 and CBX6, removes K48-linked polyubiquitin and protects them from proteasomal degradation. Knockdown of Usp26 increases AP+ iPSC colony generation fourfold.

KEY FINDING 1 OF 5

claim	shRNA knockdown of Usp26 enhances reprogramming efficiency fourfold
protein_pair	USP26 ↔ NANOG
evidence	~400 AP+ colonies vs ~100 in control shRNA (quantitative)
context	Dox-induced OSKM MEF reprogramming, day 12
confidence	0.95
source_span	Page 3, Para 1

KEY FINDING 3 OF 5

protein_pair	USP26 ↔ CBX6
residues	Cys304
evidence	Loss of K48 deubiquitination in C304S and Δ295–312 mutants (qualitative)
source_span	Page 5, Para 1

Fingerprints are embedded (PubMedBERT) for semantic search and aggregated into the interaction graph. Three complete examples: Supplemental Information 4.

SCOPE AND AVAILABILITY

In silico results, at low cost.

	CALCRL / RAMP1	PD-1 / PD-L1	PVR / TIGIT
Pass all gates, ipTM ≥ 0.7	17.9%	9.6%	10%
LLM API cost	$0.33	$0.42	$0.39
GPU time (local)	~11 h	~34 h	~20 h

The top 20 designs of each campaign were refolded with Boltz-2 and Protenix-v2; the majority pass all gates in all folding models.

LIMITATIONS STATED IN THE MANUSCRIPT

Experimental success rates have not been evaluated; in silico metrics are a proxy. The database (~14,000 papers in the example runs) limits target choice, and the released database is smaller because of licensing.

AVAILABILITY

Manuscript with Supplemental Information 1–6, appendices, code, skills and corpus tooling:

`github.com/MichaUckelmann/little-protein-tiger`